

THE DIGITAL SUBSURFACE: DEMYSTIFYING RESERVOIR SIMULATION

A comprehensive guide to the physics, mathematics, and models powering modern reservoir engineering.

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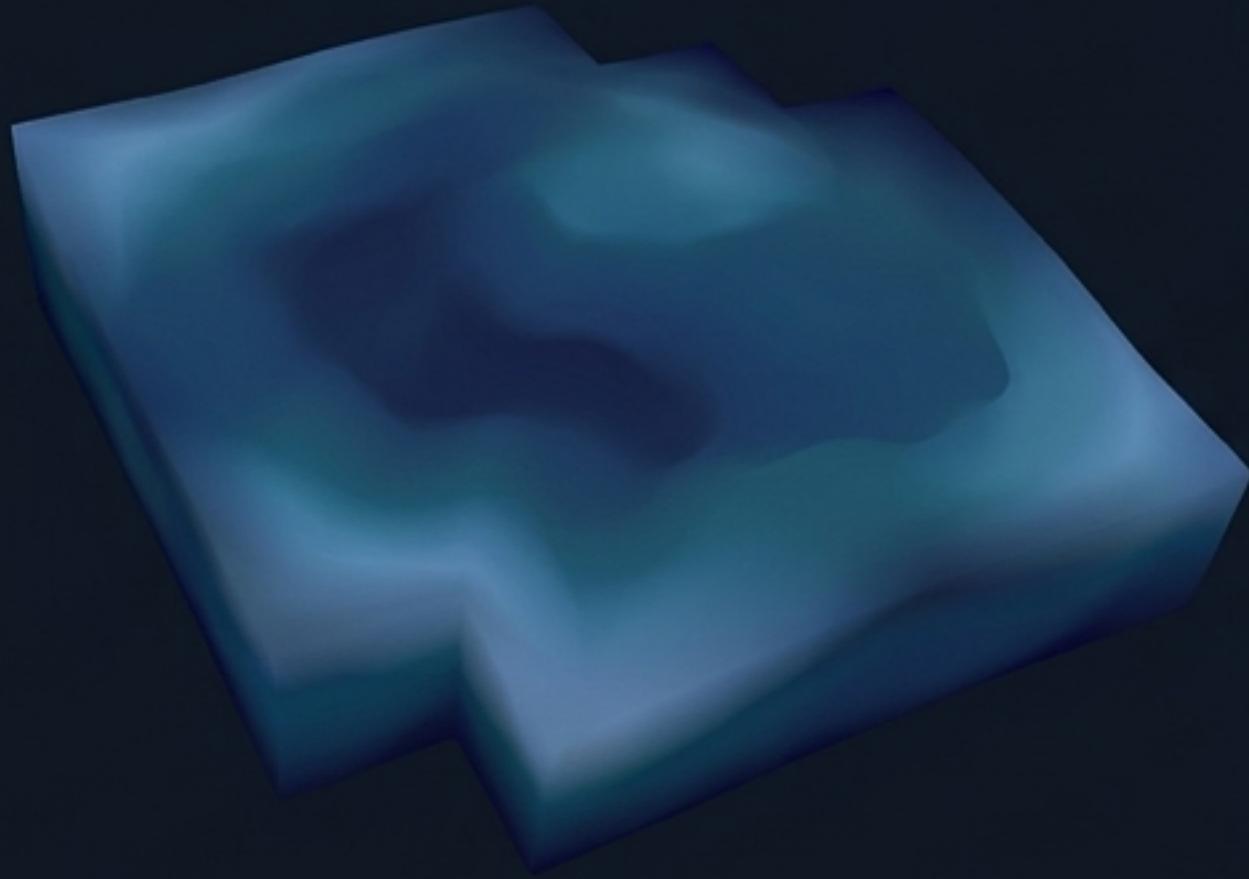
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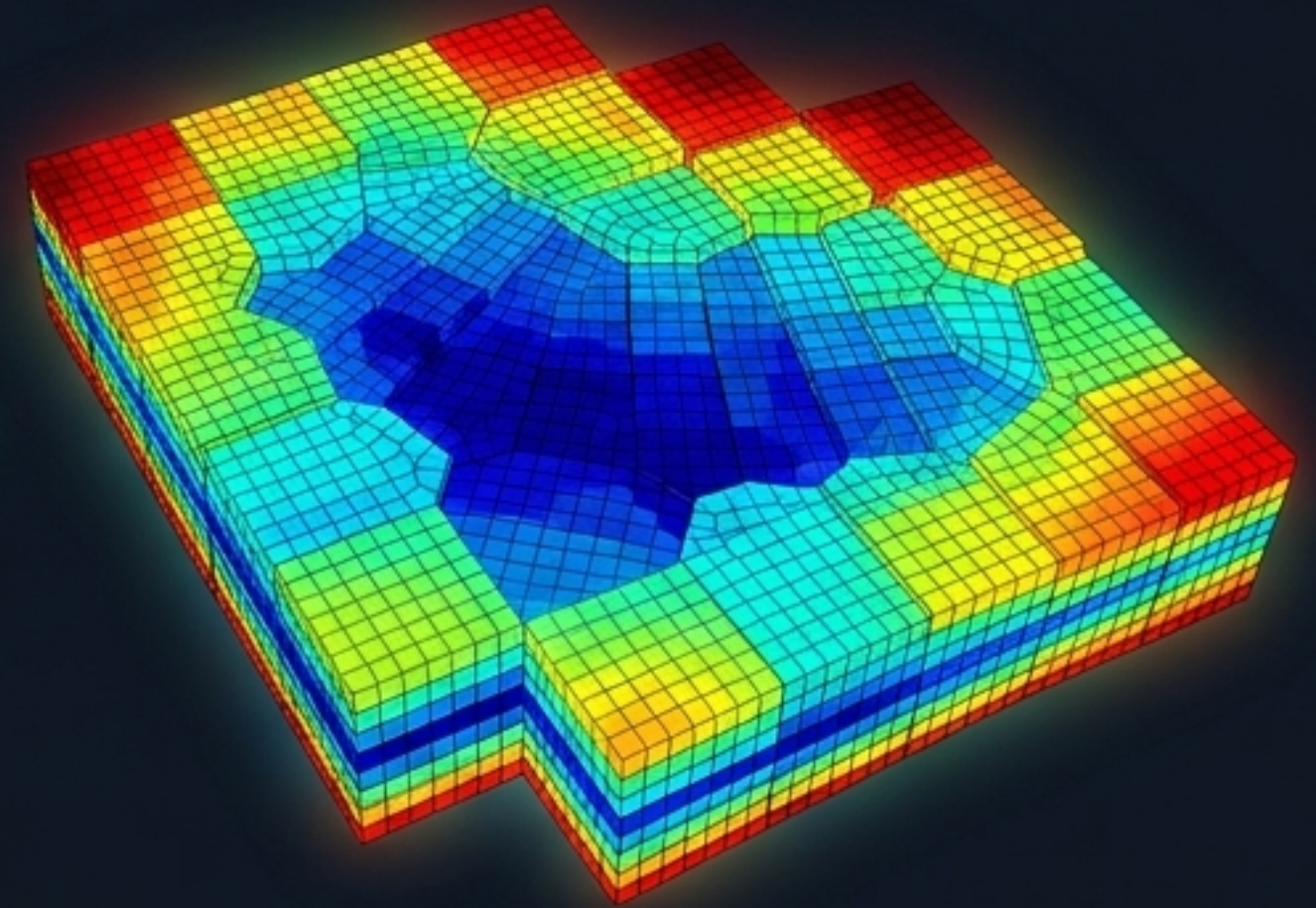
The Paradigm Shift: From Continuous to Discrete

Reservoir simulation is the representation of petroleum reservoir performance using a mathematical model. Instead of applying conventional theories to the entire reservoir as a single unit, we apply them to discrete, smaller parts called grid blocks.

Conventional View (Macroscopic)

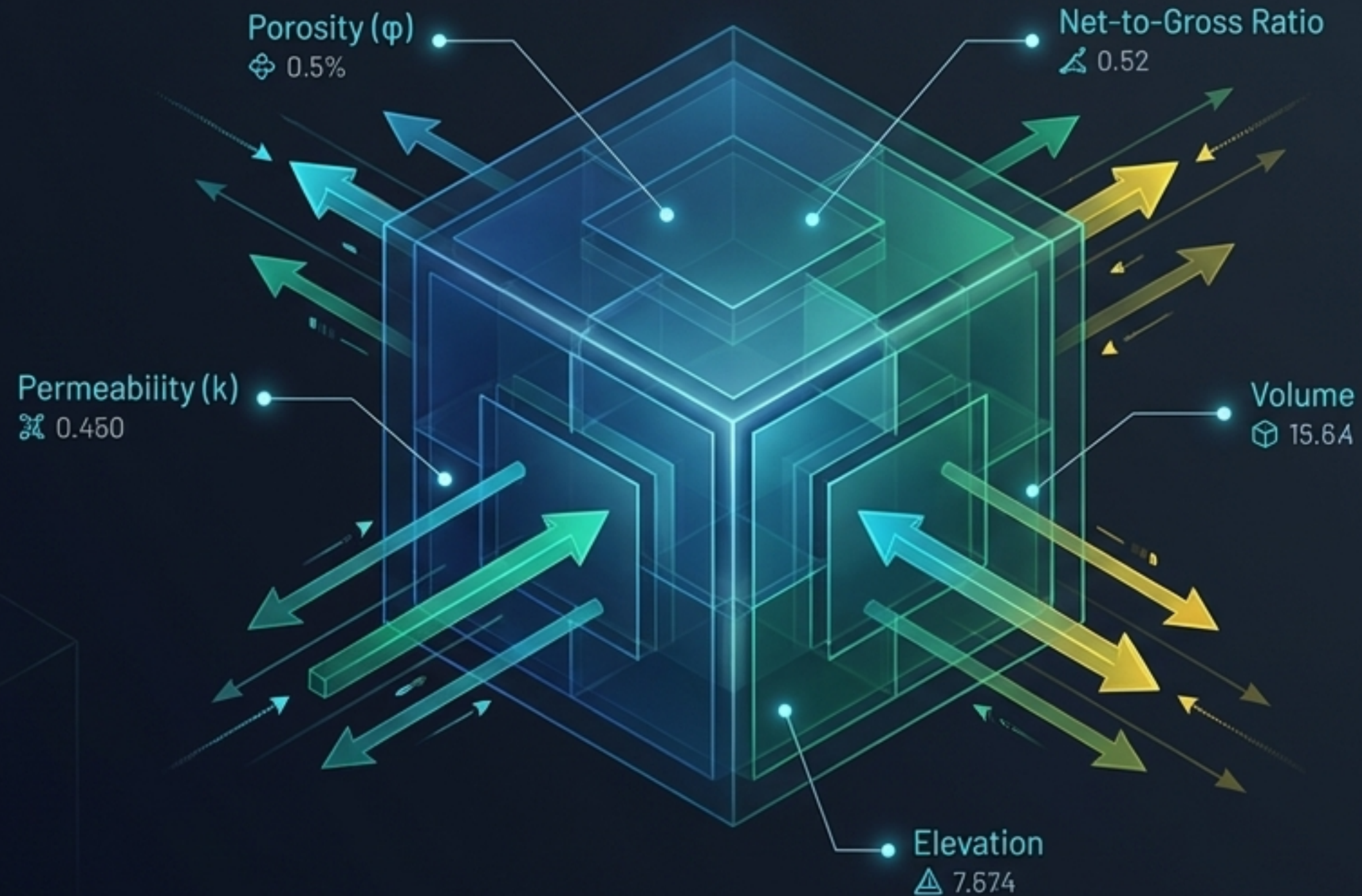


Simulation View (Microscopic)



The Grid Block Microcosm

Every single grid block operates as its own small reservoir. The simulation calculates the volume of each phase (oil, gas, water) as time progresses, driven by pressure and fluid movement between neighboring blocks.



The **Three Pillars** of the Simulation Engine

Material Balance



Known as the continuity equation or conservation of mass. Ensures no matter is lost.



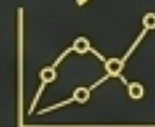
Darcy's Law



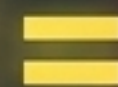
The fundamental fluid flow equation dictating how fluids move through porous rock.



Equation of State (EOS)



Determines the PVT (Pressure-Volume-Temperature) properties, or handled via PVT tables.



Governing Simulation Equations

The physical reality of every single grid block, digitized.



Translating Physics into Mathematics

Combining the three pillars yields three specific, non-linear equations for every grid block—one for each phase.

Oil

$$\Delta(a_o^{n+1} \Delta p^{n+1}) = \frac{1}{\Delta t} \left[\left(\frac{V_p S_o}{B_o} \right)^{n+1} - \left(\frac{V_p S_o}{B_o} \right)^n \right]$$

Water

$$\Delta(a_w^{n+1} \Delta p^{n+1}) = \frac{1}{\Delta t} \left[\left(\frac{V_p S_w}{B_w} \right)^{n+1} - \left(\frac{V_p S_w}{B_w} \right)^n \right]$$

Gas

$$\Delta(a_g^{n+1} \Delta p^{n+1}) = \frac{1}{\Delta t} \left[\left(\frac{V_p S_g}{B_g} \right)^{n+1} - \left(\frac{V_p S_g}{B_g} \right)^n \right]$$

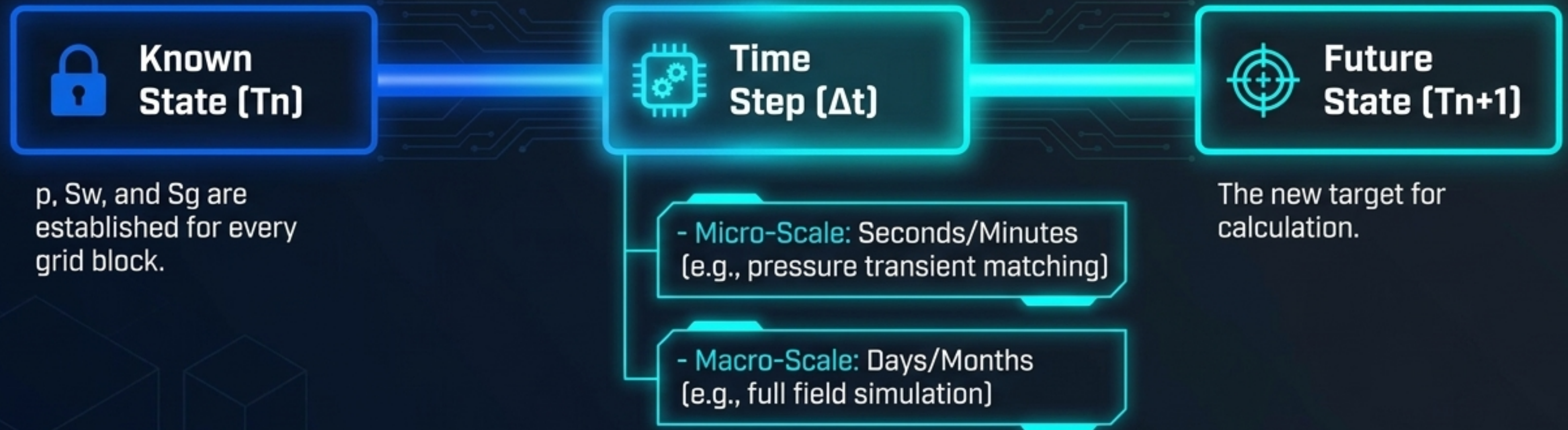
The Core Unknowns

1. Pressure (p)
2. Water Saturation (Sw)
3. Gas Saturation (Sg)

Note: "n" refers to the old time level (known quantities).

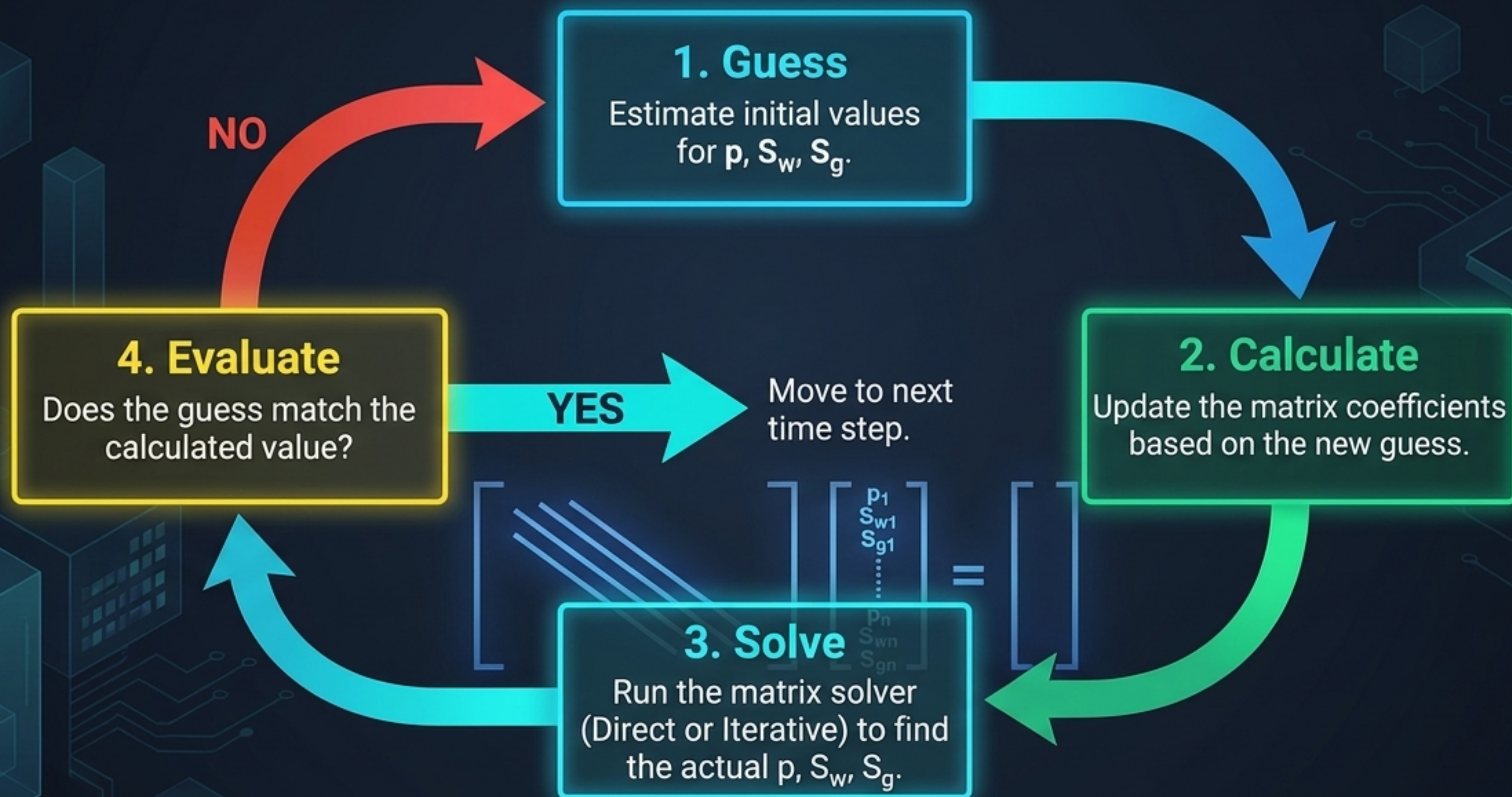
The Dimension of Time: Advancing the State

The computational challenge: advancing from a known solution to calculate variables for the exact new time level.



The Iteration Loop: Solving Non-Linear Matrices.

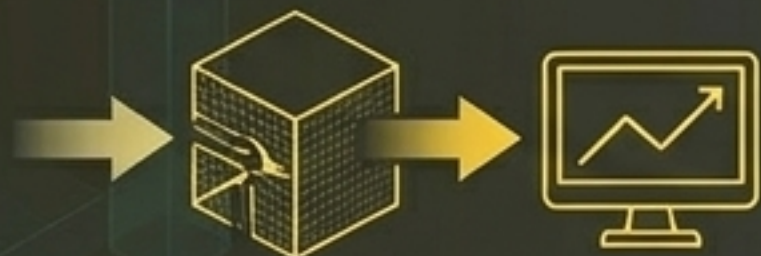
Because equations are non-linear, coefficients depend on the very variables ($\mathbf{p}$, $\mathbf{S}_w$, $\mathbf{S}_g$) we must solve for.



The Tools of the Trade: Commercial Simulators

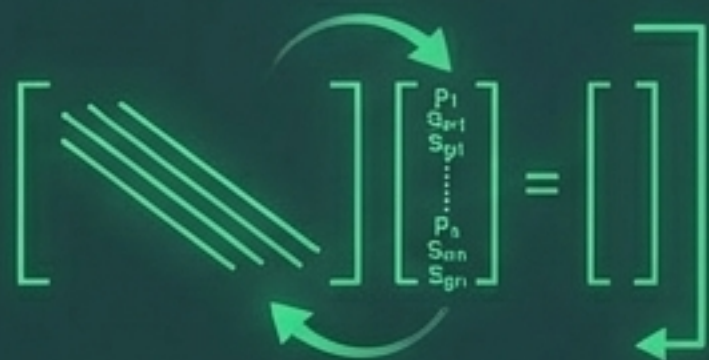
Eclipse	CMG	STAR	IMEX
GEM	TNavigator	UTCHEM	MORE

Where Simulators Differ



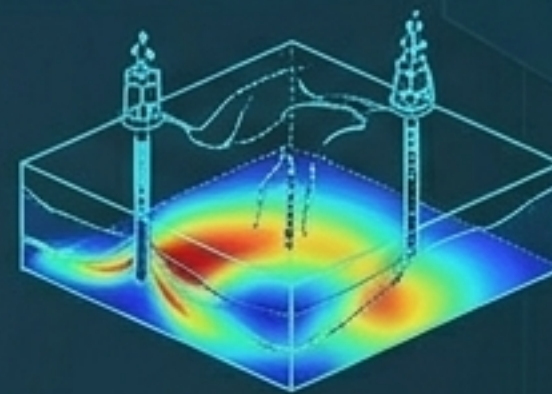
Input / Output

Pre-processing (data handling) and post-processing (visuals and reporting) capabilities vary significantly in user experience.



Matrix Solvers

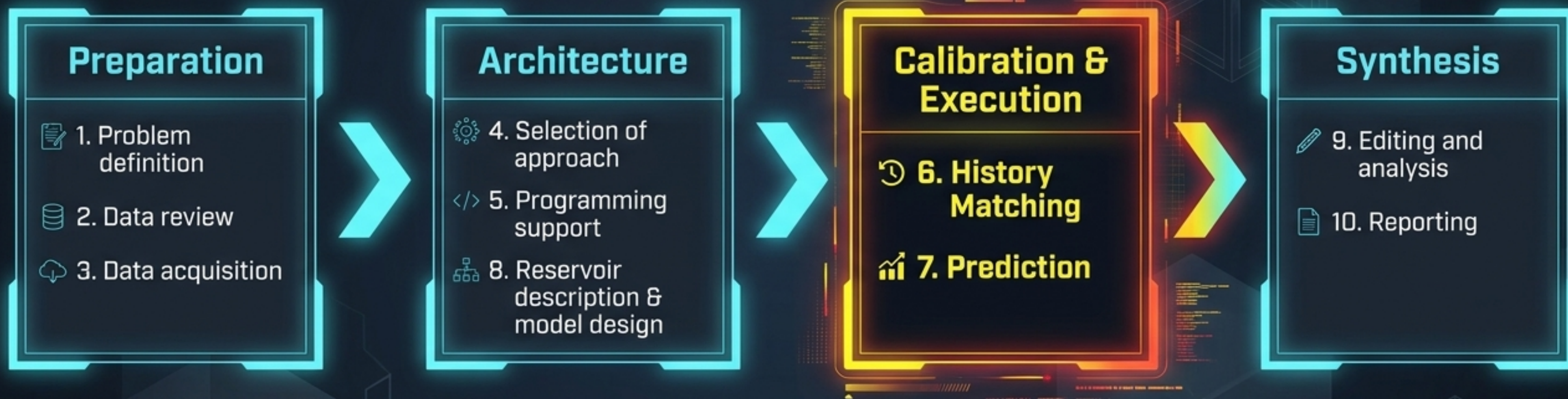
The core computational engine. Because solving matrices consumes the vast majority of computing time, developers ruthlessly optimize solver efficiency.



Specialized Applications

Versatility for unique physical conditions, such as modeling complex Enhanced Oil Recovery (EOR) processes.

The Lifecycle of a Simulation Study

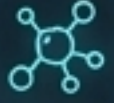


Part IV: Model Typologies

Selecting the right digital architecture for the physical reality.

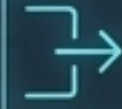
- 01. The Black Oil Model
- 02. The Compositional Model
- 03. The Chemical Flood Model
- 04. Thermal Models
- 05. Dual-Porosity Models

The Black Oil Model: The Industry Standard



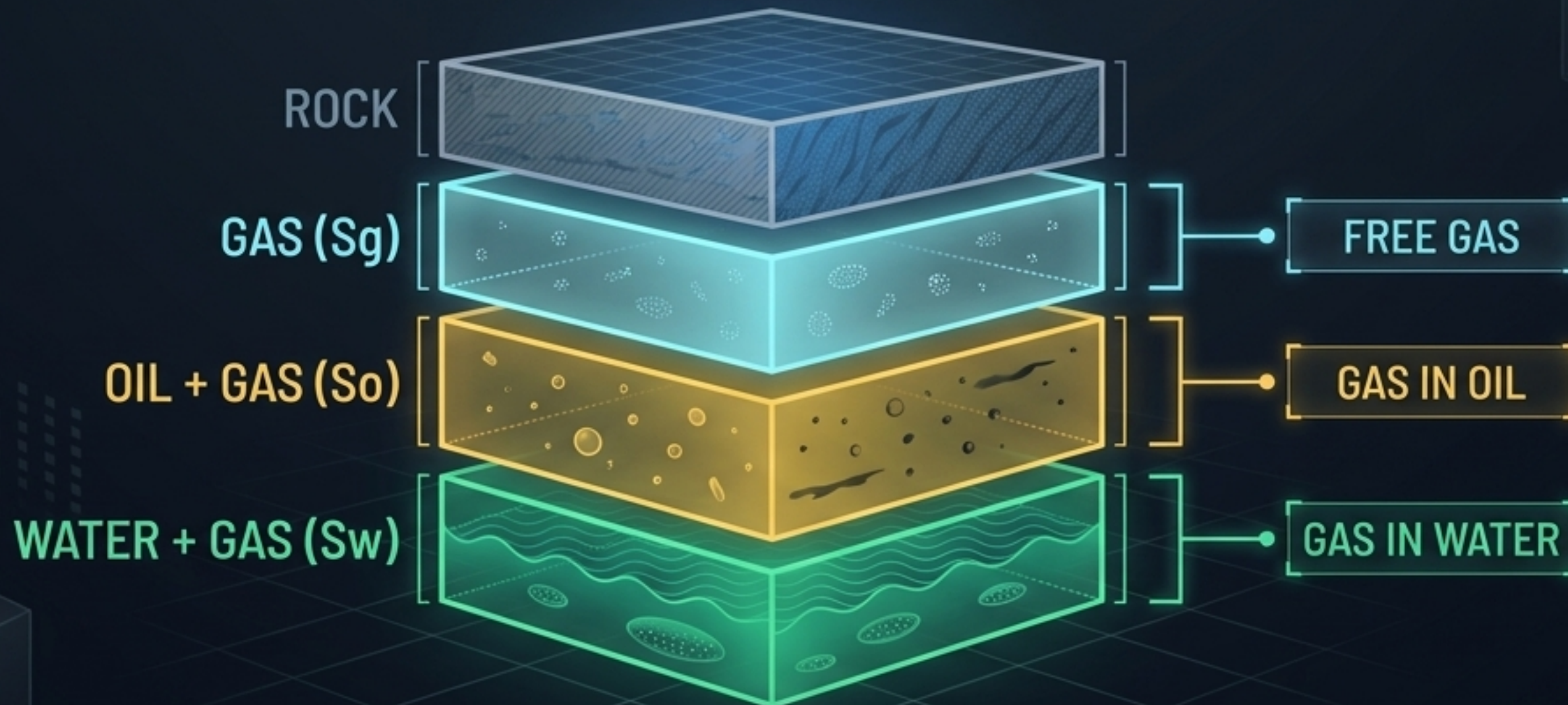
CORE CONCEPT

- Treats oil, gas, and water as mass components.
- Crucially, gas is allowed to dissolve in both the oil and water phases, tracked via solubility factors (R_{so} and R_{sw}).



TARGET PROCESSES

- Primary depletion
- Waterflooding
- Immiscible gas injection
- WAG
- Capillary imbibition

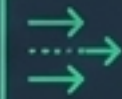


The Compositional Model: Tracking Molecular Complexity



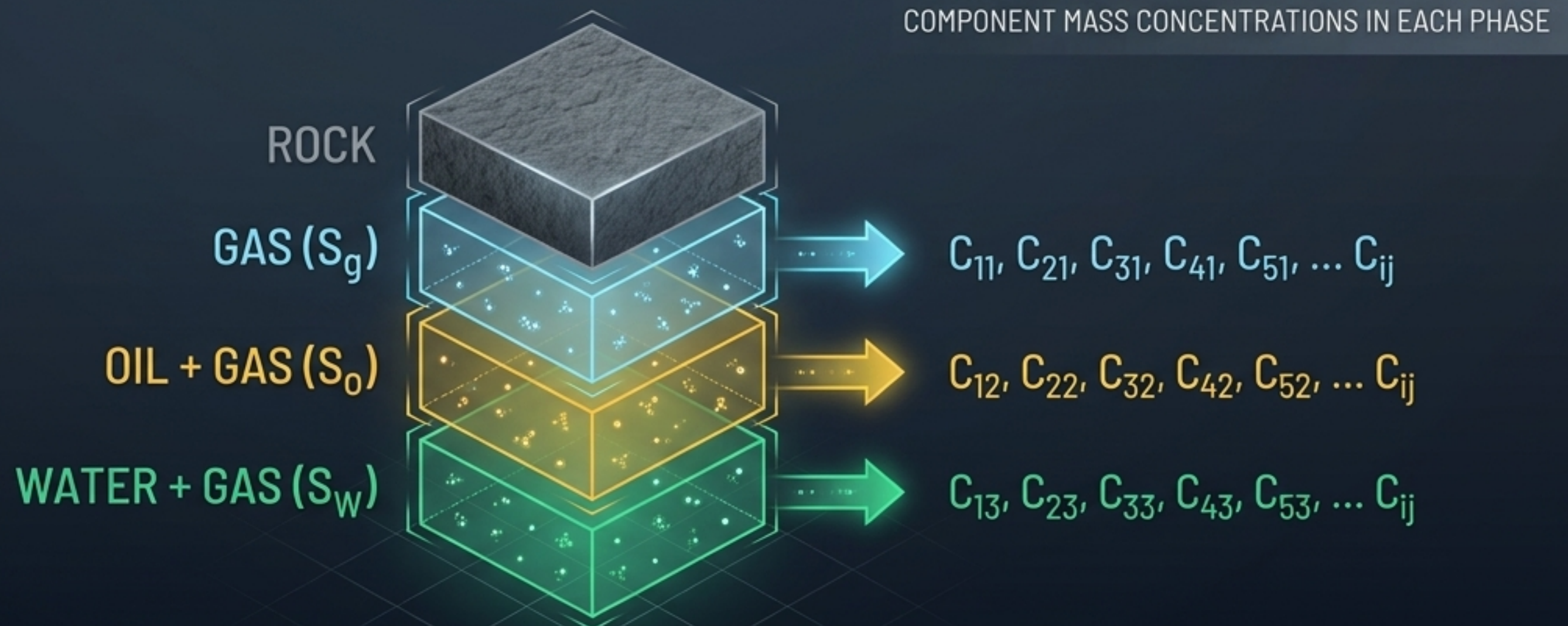
CORE CONCEPT

Acknowledges the explicit chemical compositions of the oil and gas phases due to highly complex PVT behavior that simple mass components cannot capture.



TARGET PROCESSES

- Gas injection (miscibility via CO₂ flooding)
- Near-critical reservoirs
- Gas recycling in condensate reservoirs



The Chemical Flood Model: Artificial Displacement

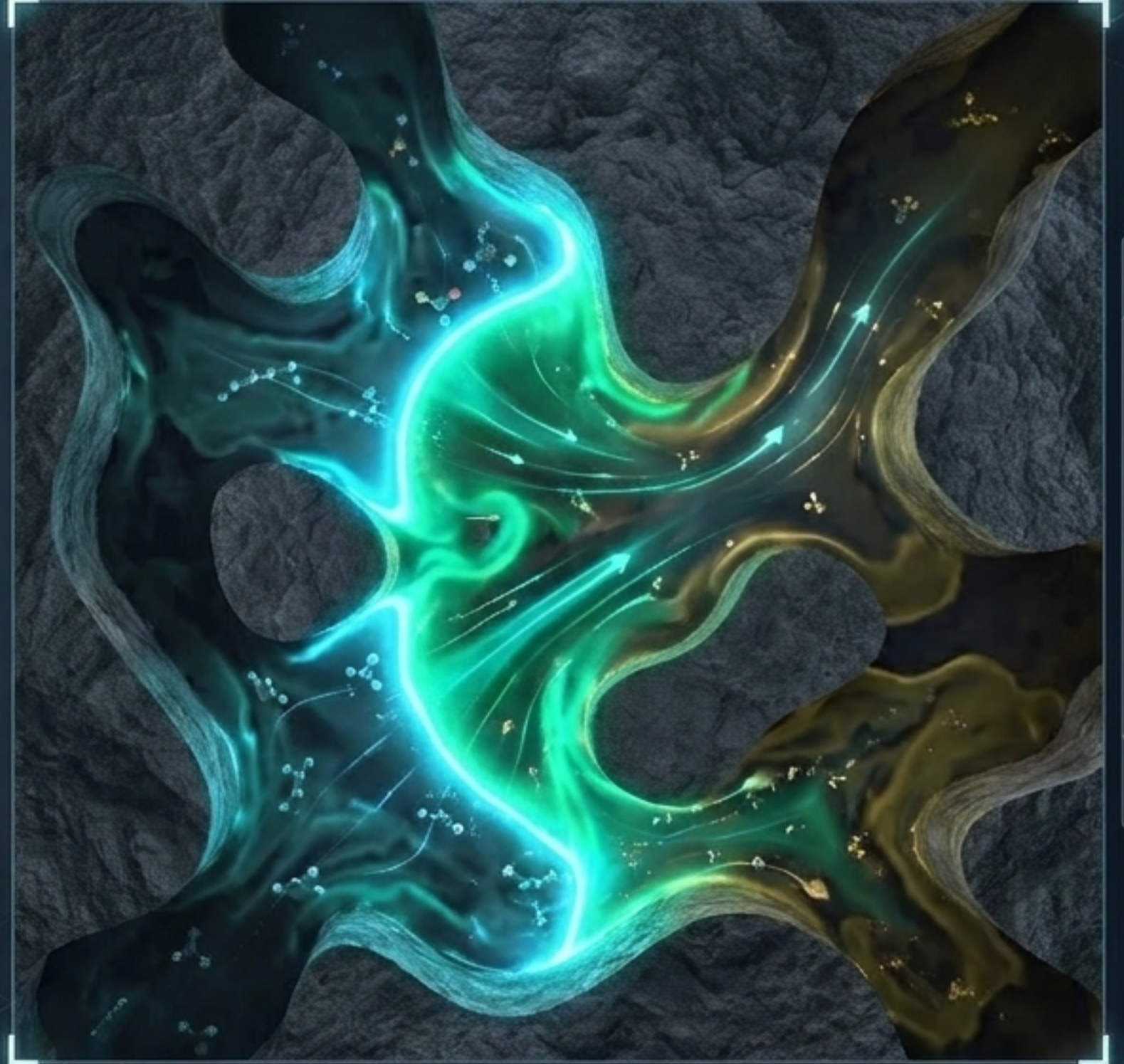


CORE CONCEPT

Simulates complex displacement processes involving injected chemicals that alter fluid interactions within the reservoir rock at the microscopic scale.

TARGET PROCESSES

- Polymer flooding
- Surfactant flooding
- Low-Tension Polymer Flooding (LTPF)
- Alkali flooding
- Foam flooding



Thermal Models: Viscosity Manipulation

CORE CONCEPT

Used for heavy oil with extremely high initial viscosity (100s to 1000s of centipoise). Simulates massive heat addition to reduce viscosity and mobilize the oil.

TARGET PROCESSES

- Steam "soaks" (Huff n' Puff)
- Steam "drive" (Flooding)
- In-situ combustion (Air injection)



THERMAL VISCOSITY REDUCTION & MOBILIZATION

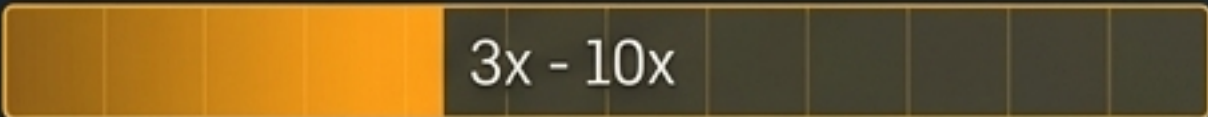
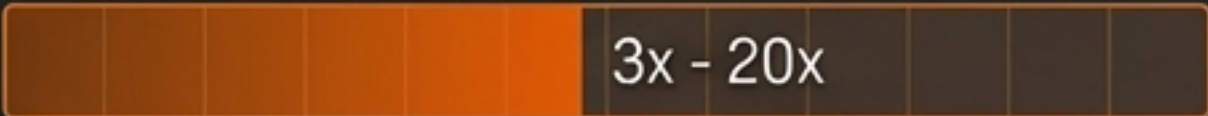
Dual-Porosity Models: Managing Fractured Systems

Designed for naturally fractured reservoirs. Simulates multiphase flow by tracking the constant exchange of fluids between two distinct domains.



Model Selection Diagnostic Matrix

Balancing physical process requirements against extreme variations in relative computing costs.

Simulator Type	Target Processes	Degree of Difficulty	Relative Computing Cost
Black Oil	Primary, Waterflood	Routine	 1x (Cheap) 1
Chemical (Polymer)	Polymer flooding	Not too difficult	 2x - 5x
Thermal (Steam)	Steam soak/flood	Not too difficult	 3x - 10x
Compositional	Gas injection/WAG	Difficult/Specialized	 3x - 20x
Thermal (In Situ)	In situ combustion	Very difficult	 10x - 40x